Machine Learning 2026 — Assignment 1

Question 4: Regression Report

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Summary. The dataset contains 873 rows across 291 groups. Linear Regression reaches validation $R^2 = 0.7543$. Ridge and Lasso are effectively tied with the baseline; Elastic Net is slightly lower. Histogram Gradient Boosting reaches validation $R^2 = 0.8802$ on the same split, with a larger train–validation gap. These are internal validation results, not hidden-test scores.

Scope note. The report follows the supplied notebook results and the assignment prompts. The actual-versus-predicted scatter plot has been removed. Graphs are colored; headings, text and tables remain plain black.

I used AI assistance to generate this report PDF; the graphs and findings are my own, produced through code I wrote, and I take full responsibility for the content to the best of my knowledge.

(b) Exploratory Data Analysis

The raw data has 22 columns: row_id, group_id, F1–F19 and target. There are 16 numerical predictors and three categorical predictors (F13, F14, F15). There are 291 groups with exactly three rows each. F3, F9, F10, F14 and F15 are constant within a group.

The target mean is 754.99, standard deviation 638.05, median 595, and range 0–3285. The mean being higher than the median indicates right-skewness. A few high targets may therefore be difficult to predict accurately.

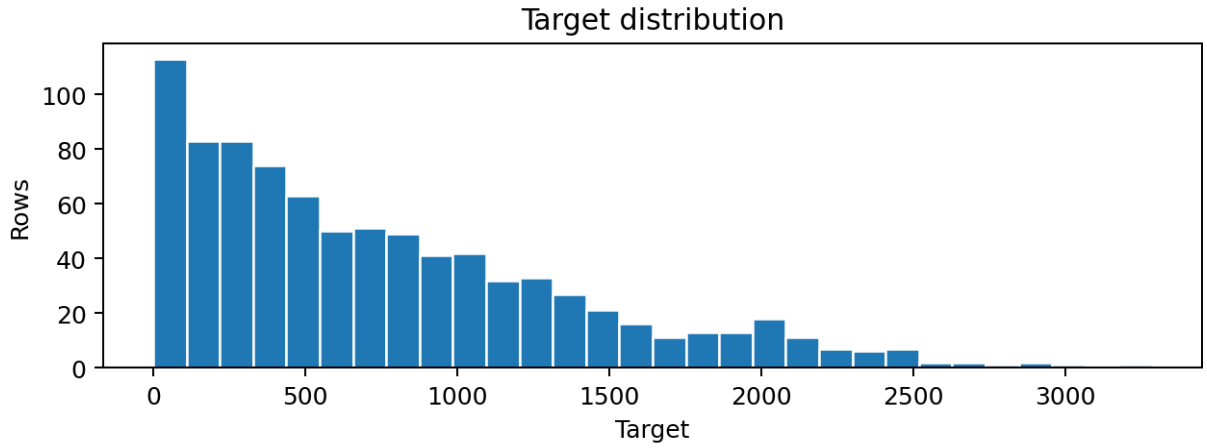


Figure 1. Target distribution: most values are lower, with a long right tail.

Missing values. 97 rows (11.1%) contain at least one missing value. No column has more than 1.83% missingness. The most-missing columns are F12 (16), F8 (14), F16 and F1 (13 each).

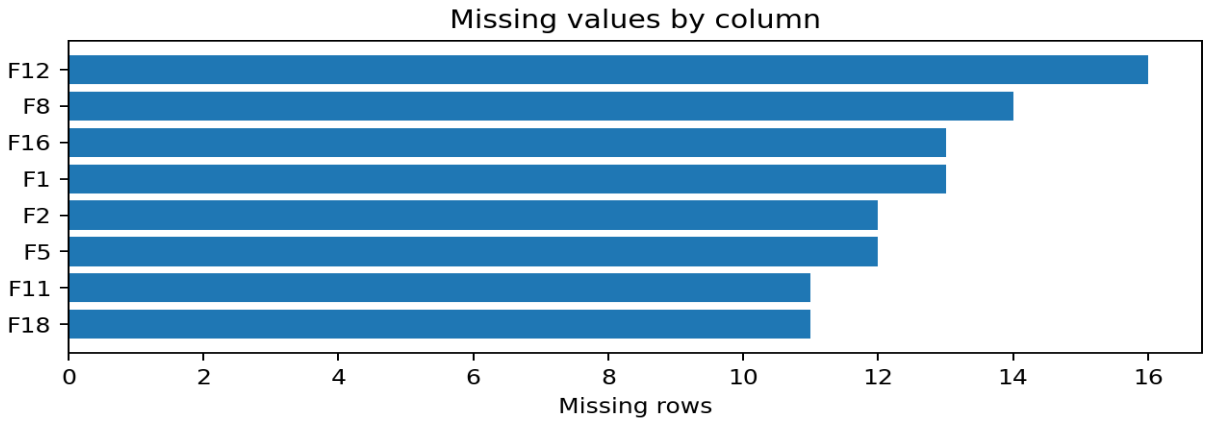


Figure 2. Missing values by column.

(b) EDA — feature relationships

The strongest numerical Pearson correlations with target are F19 (+0.471), F4 (–0.458), F16 (+0.339), F12 (+0.272) and F11 (–0.226). Correlation describes linear association only; it does not establish causation.

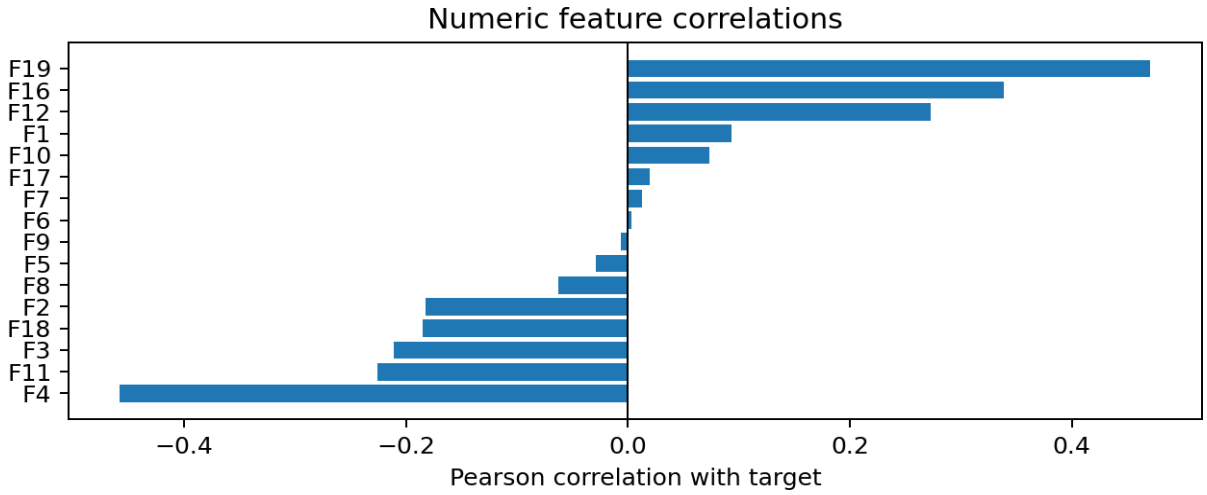


Figure 3. Pearson correlations between numeric predictors and target.

The mean target changes nonlinearly across F10: it rises toward values 6–8 and then falls. Target means also differ across the categories of F13, F14 and F15. These patterns suggest that a simple additive linear model may miss some structure, but the summaries alone do not prove cause and effect.

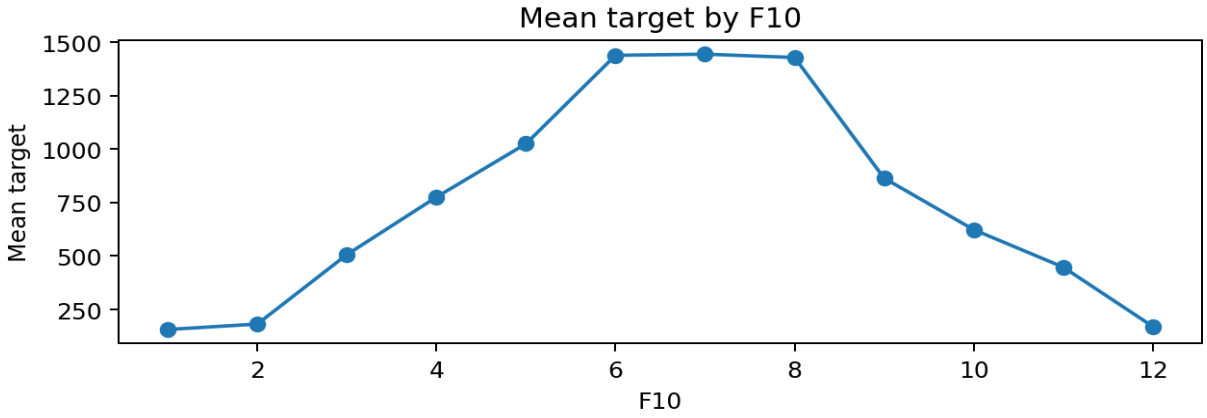


Figure 4. Mean target by F10; the curved pattern motivates testing nonlinear F10 features.

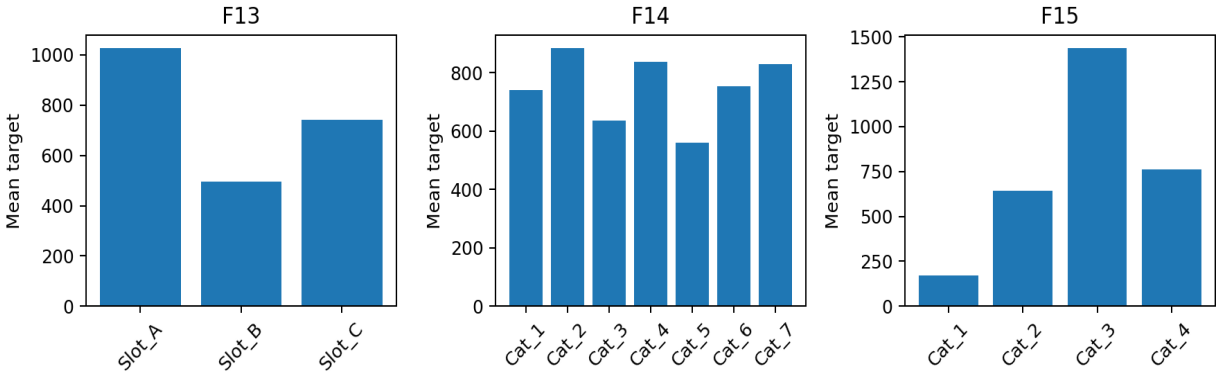


Figure 5. Mean target by category for F13, F14 and F15.

(c) Data Preparation and Feature Engineering

The notebook uses the following preparation:

• Excludes row_id and group_id from predictors; target is the label. group_id is retained for group-aware splitting and diagnostics.

• Imputes numerical missing values with the median and adds missingness indicators.

• Imputes categorical missing values with the most frequent category, then one-hot encodes F13, F14 and F15. Unknown categories are ignored by the encoder so transformation does not fail.

• Produces 38 features: 16 numerical values, 8 missingness indicators and 14 one-hot indicators. The notebook checks transformed values for NaNs and checks that identifiers are excluded.

Appropriateness and limitation. This approach keeps rows despite modest missingness and converts categories into model-ready columns. However, the main preprocessing is fitted before the train/validation split, so validation rows influence imputation statistics and the learned category vocabulary. This is feature-distribution leakage, not target leakage. A stricter evaluation fits preprocessing on training groups only.

(d) Validation Strategy

GroupShuffleSplit uses test_size = 0.20 and random_state = 2024514. It produces 696 training rows and 177 validation rows. Whole groups stay together, which better reflects the hidden-test setting of unseen groups than a random row split. Because only one split is used, its score may depend on which groups were held out.

(e) Baseline Model and Evaluation

The baseline is ordinary Linear Regression with no tuned hyperparameters.

Metric	Training	Validation
R^2	0.7493	0.7543
RMSE	319.09	316.87
MAE	236.03	233.30

Training R^2 is 0.005 lower than validation R^2, so the baseline shows almost no train–validation gap on this split. The errors remain substantial.

(f) Diagnostics

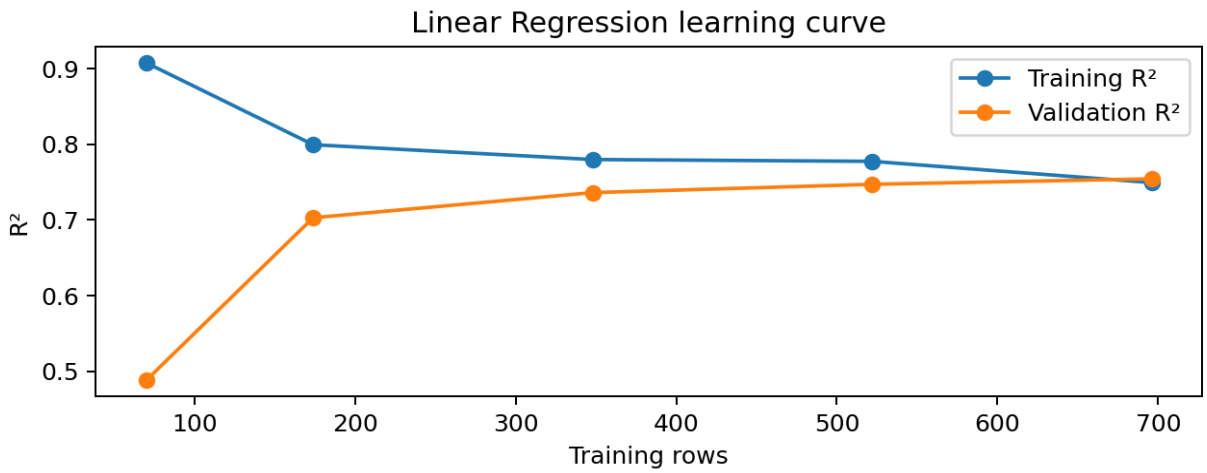


Figure 6. Validation R^2 rises as more training rows are used, while training R^2 falls. At full size the scores approach one another.

The baseline's largest errors include underprediction of high targets (for example, actual 2501 and predicted about 1428) and overprediction of lower targets (actual 650 and predicted about 1491). The residuals show a tendency toward the middle of the target range; they do not prove one cause for every error.

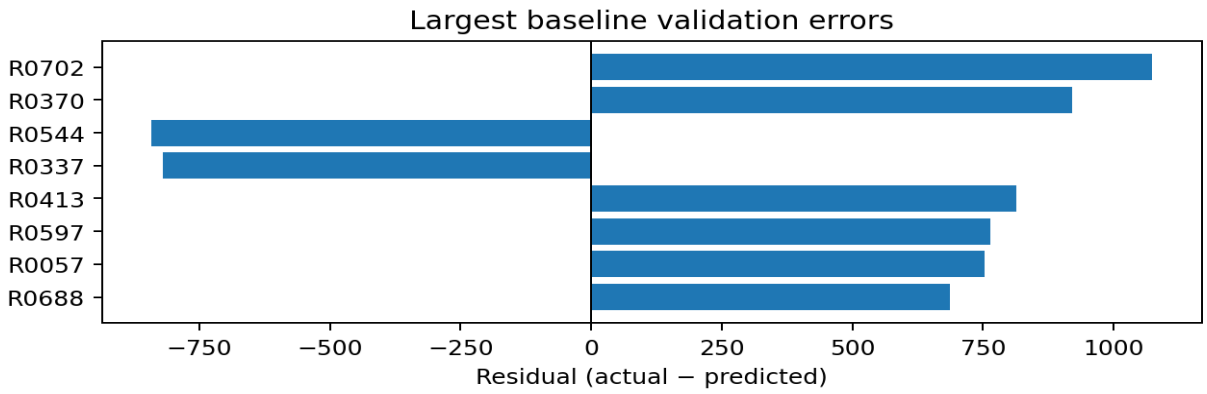


Figure 7. Largest baseline validation residuals. Positive residuals mean underprediction; negative residuals mean overprediction.

(g) Model Improvement

The alternatives use the same group-aware split. Ridge and Lasso at alpha = 0.01 perform almost identically to Linear Regression; Elastic Net is slightly lower. Histogram Gradient Boosting can represent nonlinear splits and interactions.

Model	Train R^2	Val. R^2	Val. RMSE	Val. MAE
Hist. Gradient Boosting	0.9569	0.8802	221.22	154.84
Linear Regression	0.7493	0.7543	316.87	233.30
Lasso (0.01)	0.7493	0.7543	316.87	233.30
Ridge (0.01)	0.7493	0.7543	316.88	233.31
Elastic Net	0.7455	0.7476	321.16	236.42

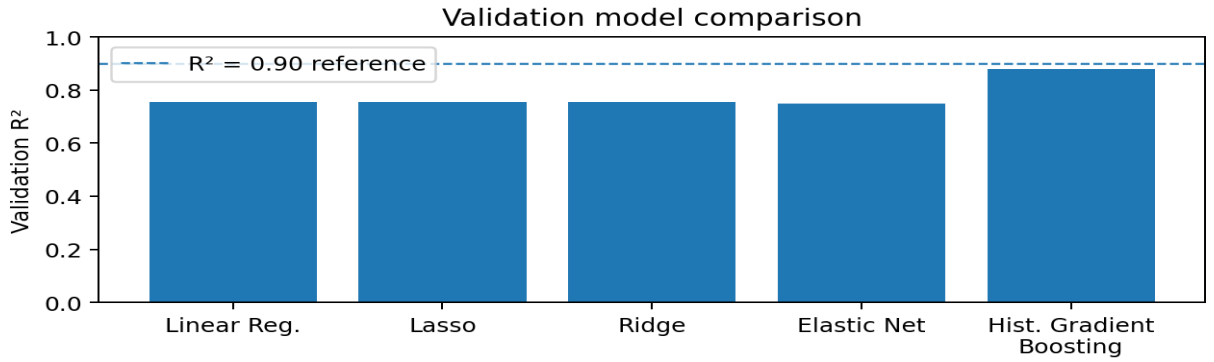


Figure 8. Validation R^2 for the tested models; 0.90 is a reference line, not a score achieved here.

Histogram Gradient Boosting improves validation R^2 from 0.7543 to 0.8802, and lowers RMSE from 316.87 to 221.22 and MAE from 233.30 to 154.84. Its training R^2 is 0.9569, so the train–validation gap is wider (about 0.0767). The score is still below 0.90 and is not a hidden-test result.

(h) Generalization and Limitations

(i) One way the pipeline could fail on unseen groups. A new group may contain feature values or target patterns that are less well represented in the training groups. The baseline then may predict values closer to the common middle range than the true values. This is consistent with observed validation errors: one row with target 2501 was predicted around 1428, while a row with target 650 was predicted around 1491. These examples show a weakness on the held-out split; they do not prove exactly how hidden groups will behave. Because group_id is excluded, the model cannot use a group-specific correction for a group it has never seen.

(ii) One change to try next. Add a small nonlinear feature for F10, such as $F10^2$, and test it with Ridge regression. This is motivated by the observed mean-target curve: it rises toward F10 values 6–8 and then falls, which a straight-line effect may not capture. To test fairly, split by group first, fit imputation and encoding only on training groups, and compare R^2, RMSE and MAE on held-out groups. Keep the change only if it improves results consistently. This is a testable idea, not a promise of R^2 above 0.90.